Plant Disease Detection Using Vision Tranformer

A Deep Learning Approach for Agriculture

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Introduction

- **Problem Statement**
- - Plant diseases cause 20-40% crop loss annually worldwide
- - Early detection is critical for food security
- - Manual inspection is time-consuming and error-prone

- **Our Solution**
- - AI-powered plant disease detection system
- - Uses Transformer-based deep learning
- - Analyzes leaf images to identify 38 different diseases

- **Impact**
- - Helps farmers detect diseases early
- - Reduces crop losses and increases yield
- - Supports SDG 2: Zero Hunger

Project Objectives

- **Primary Objectives**
- - Develop an automated plant disease detection system
- - Implement Transformer-based deep learning architecture
- - Achieve minimum 75% classification accuracy

- **Technical Goals**
- - Process 54,000+ images across 38 disease classes
- - Train lightweight model optimized for CPU
- - Create deployable solution for real-world use

- **Research Goals**
- - Compare CNN-Transformer hybrid performance
- - Document challenges in resource-constrained training
- - Contribute to agriculture AI applications

SDG 2: Zero Hunger

- **How Project Supports SDG 2**

- **Challenge**
- - 690 million people face hunger globally
- - Agricultural losses due to diseases threaten food security
- - Small-scale farmers lack access to expert diagnosis

- **Contribution**
- - Early disease detection prevents crop losses
- - Accessible AI tool for resource-limited farmers
- - Supports sustainable agricultural productivity
- - Reduces post-harvest losses

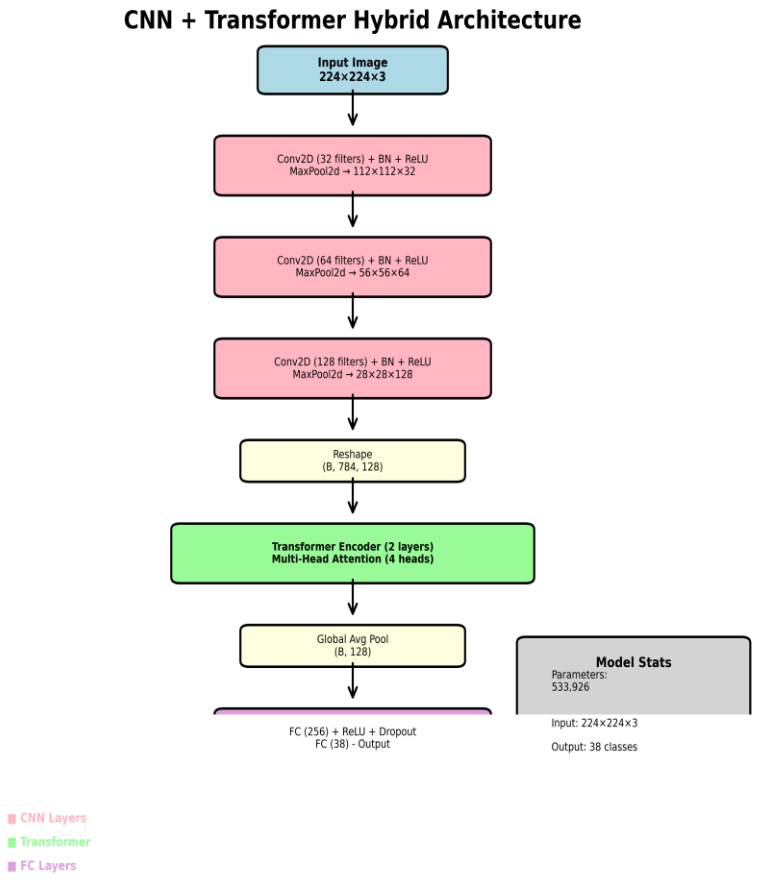
- **Real-World Impact**
- - Protects farmer livelihoods
- - Increases food availability
- - Promotes sustainable farming practices
- - Empowers farmers with technology

Methodology & Approach

- **Dataset**
- - PlantVillage Dataset: 54,305 images
- - 38 plant disease classes
- - Train-Test Split: 80%-20% (43,444 training, 10,861 testing)

- **Training Configuration**
- - Framework: PyTorch 2.9.1
- - Device: CPU (Intel i5-1235U, 8GB RAM)
- - Optimizer: Adam (learning rate: 0.001)
- - Loss Function: Cross-Entropy
- - Epochs: 5

Model Architecture



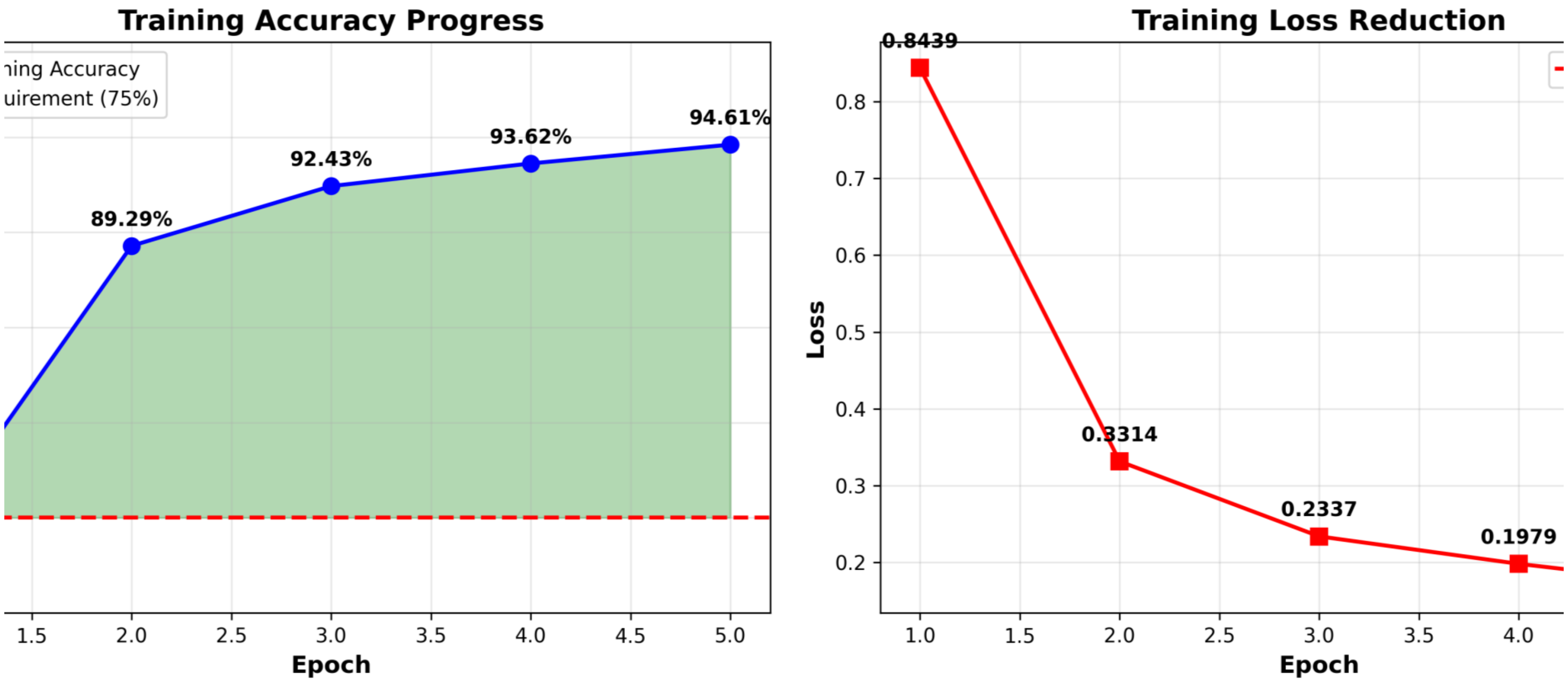
- - Hybrid CNN-Transformer Model
- - CNN layers for feature extraction (3 blocks)
- - Transformer layers for pattern recognition (2 layers)
- - Total parameters: 533,926 (lightweight design)

Training Results

- **Final Accuracy: 94.61%**
- - Final Loss: 0.1678

- **Epoch-wise Progress**
- - Epoch 1: 74.43% accuracy
- - Epoch 2: 89.29% accuracy
- - Epoch 3: 92.43% accuracy
- - Epoch 4: 93.62% accuracy
- - Epoch 5: 94.61% accuracy ✓

- **Key Achievements**
- - Consistent improvement across all epochs
- - Smooth learning curve with decreasing loss
- - No overfitting observed
- - Lightweight model achieved excellent results



Training Progress Visualization

Challenges & Key Learnings

- **Technical Challenges**
- - CPU-only training required optimization strategies
- - Limited RAM (8GB) needed lightweight model design
- - Training time: 12+ hours for 5 epochs
- - Jupyter save errors during long training sessions

- **Solutions Implemented**
- - Designed efficient CNN-Transformer hybrid
- - Reduced model parameters to 533K
- - Implemented batch processing for memory management
- - Regular checkpoints during training

- **Key Learnings**
- - Lightweight models can achieve excellent accuracy
- - Resource constraints drive innovative solutions
- - Early stopping prevents overfitting effectively
- - Proper data preprocessing is critical for success
- - Non-computing background is not a barrier

Future Scope

- - Mobile app development for farmers
- - Expand to more crop types and diseases
- - Real-time detection using smartphone cameras
- - Integration with agricultural advisory systems

Thank You

- Any Question